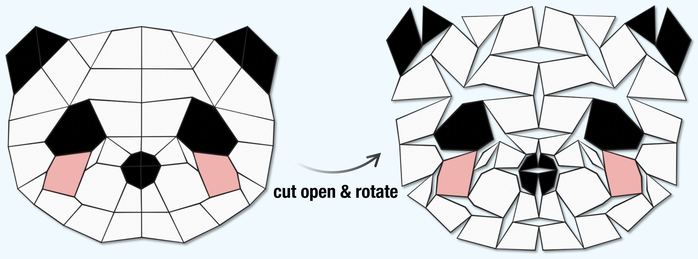
Uniformly Deployable Kirigami on Arbitrary Planar Graphs

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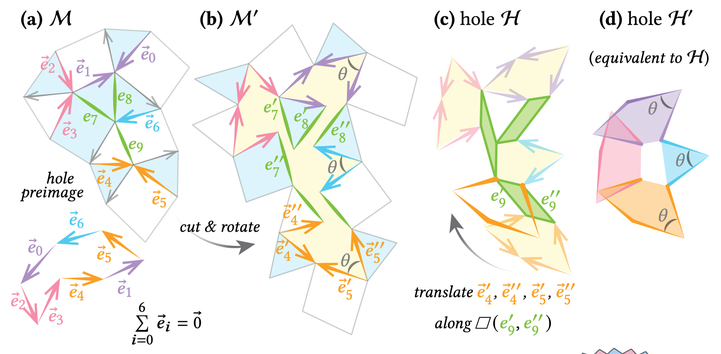
Tutte Auxetic Embedding Problem

Given a planar graph $\mathcal{G} = (\mathcal{V}, \mathcal{F})$, does it exist an embedding X such that the resulting kirigami structure after cutting the mesh $(X, \mathcal{F})$ open, is uniformly deployable?



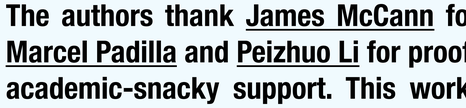
Hole analysis

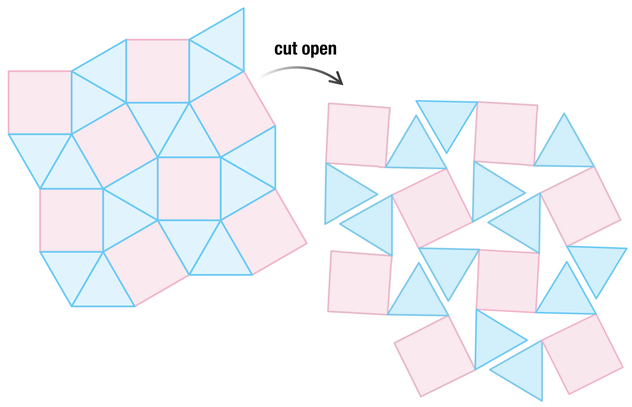
Uniform deployability of the resulting kirigami structure after cutting can be equivalently formulated as a linear constraint on the embedding of the original tiling before cutting.



Definitions of cutting operations

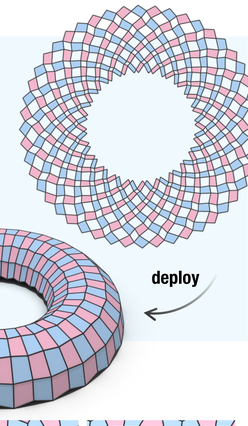
- ❖ split cut → adjacent faces with the same orientation; after cutting, completely separated
- ❖ hinge cut → adjacent faces with opposite orientations; after cutting remained hinged



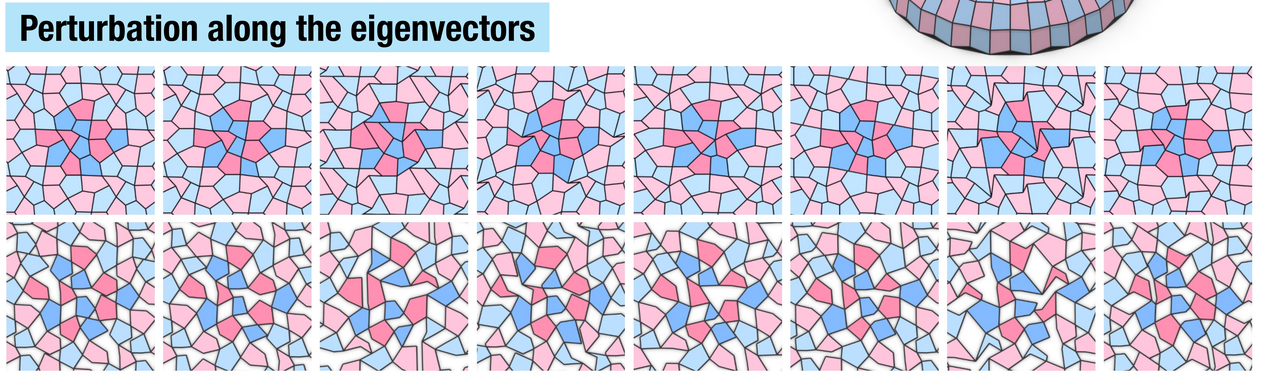


Full space of Tutte auxetic embeddings

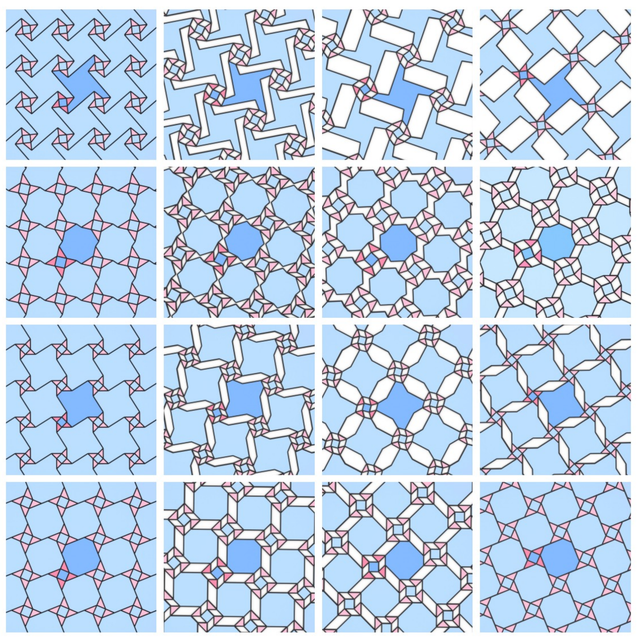
- ❖ Tutte auxetic embeddings can be solved via a linear system
- ❖ The null space of the linear system defines all feasible Tutte auxetic embeddings
- ❖ Offsetting the eigenvectors of the null space allows easy navigation (see below)
- ❖ This formulation allows pattern optimization within the design space
- ❖ Also allows inverse design for shape approximation



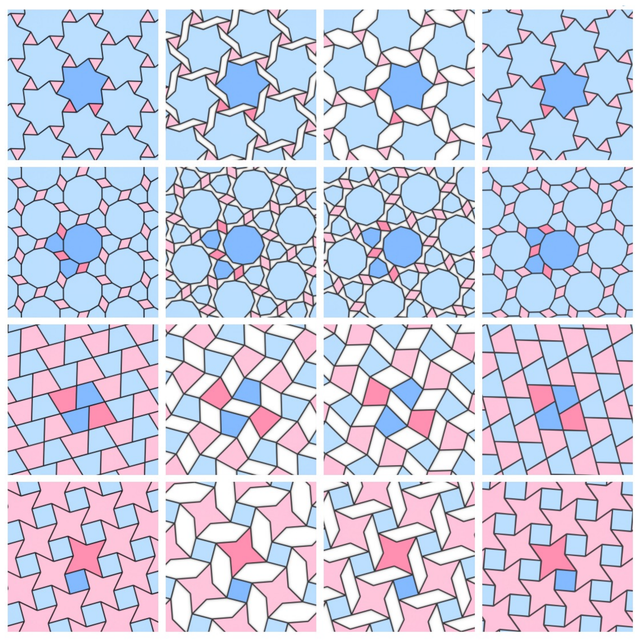
Perturbation along the eigenvectors



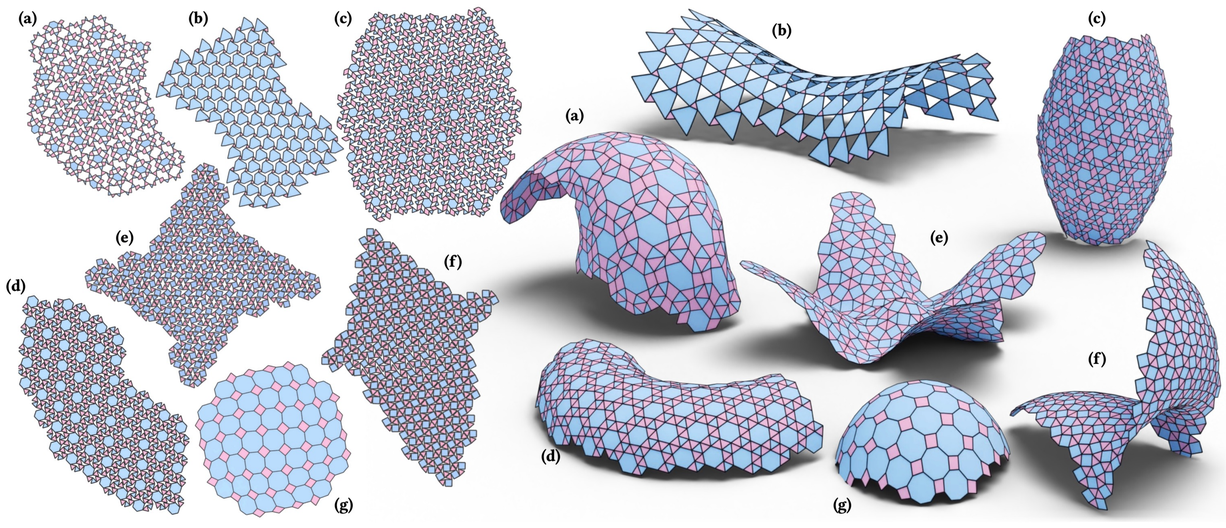
Isotropic patterns (conformal deployment)



closed at fully deployed state



Results: inverse design



Acknowledgement The authors thank the anonymous reviewers for their valuable feedback. The authors thank James McCann for his inspiring questions. Special thanks to Ruben Wiersma, Marcel Padilla and Peizhuo Li for proofreading. The authors thank all IGL members for their spiritual-academic-snacky support. This work was supported in part by the ERC Consolidator Grant No. 101003104 (MYCLOTH).